A Projection Filtering Method for Output Redundant Systems

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Abstract

Cum M. Cicero consul Nonis Decembribus senatum in aede Iovis Statoris consuleret, quid de iis coniurationis Catilinae sociis fieri placeret, qui in custodiam traditi essent, factum est, ut duae potissimum sententiae proponerentur, una D. Silani consulis designati, qui morte multandos illos censebat, altera C. Caesaris, qui illos publicatis bonis per municipia Italiae distribuendos ac vinculis sempiternis tenendos existimabat.

Key words: Cicero; Catiline; orations.

1 Introduction

Gauge invariance is a central organizing principle of modern physics, expressing the idea that observable quantities should remain unchanged under certain local transformations of the mathematical description. In classical field theories and quantum electrodynamics, this concept ensures that different representations related by gauge transformations correspond to the same physical reality [5],[7]. From an abstract viewpoint, gauge invariance provides a systematic way to separate physically meaningful dynamics from redundant coordinate or potential offsets. This symmetry-based perspective has proven essential for constructing robust and consistent models in physics, and it motivates the extension of invariance principles beyond their traditional domain toward engineered dynamical systems subject to measurement distortions and unknown offsets.

In linear control systems, similar representation issues arise in the presence of redundant measurements or sensing architectures with non-minimal output representations. In particular, when the output matrix does not have full row rank, multiple output vectors may correspond to the same underlying state. This situation naturally appears in multi-sensor configurations, distributed systems, and sparse measurement settings, where the dimension of the measurement space exceeds the effective information content [1],[3]. As a result, the output representation contains directions that do not contribute to state observability but nevertheless affect the estimation process.

From this perspective, the measurement space can be decomposed into physically meaningful components and redundant directions that do not affect the underlying state. However, conventional observer designs typically treat all output components uniformly, without explicitly distinguishing between these subspaces. This may lead to estimation dynamics that are influenced by representation-dependent artifacts, rather than purely by physically relevant discrepancies.

This observation suggests that the estimation process should be constructed in a way that isolates the physically relevant part of the output from its redundant representation. A natural mechanism to achieve this separation is through a projection operator acting on the measurement space, which removes components lying in the non-informative directions while preserving the observable content. Such projection-based constructions are also well known in physics, particularly in gauge theories, where physically equivalent configurations are related by symmetry transformations and a representative is selected via gauge fixing [5].

In this work, we propose a way of constructing a projection that acts as a static filter to recover the linear output mapping in terms of its components under given

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constraints. This projection is given as an alternative to the classical observer design for output redundant systems, due to the fact that no dynamical estimation is necessary to reconstruct states and disturbances.

The paper is organized as follows; preliminaries are given in section 2, the proposed projection filtering method is addressed in section 3. The proposed method is exemplified in section 4 with numerical examples. The conclusion and future work is given in section 5.

2 Preliminaries

2.1 The System Model

Multi output linear time invariant dynamical systems are expressed as,

$$\dot{x} = Ax + B_w w,\ y = Cx + D_w w \tag{1}$$

where $x \in \mathbb{R}^{n\times 1}$ represents the state, $w \in \mathbb{R}^{r\times 1}$ originally the bias or more generally the exogenous input and $y \in \mathbb{R}^{p\times 1}$ is the measured output. The system matrices are defined as $A \in \mathbb{R}^{n\times n}$, $B_w \in \mathbb{R}^{n\times r}$, $C \in \mathbb{R}^{p\times n}$ and $D_w \in \mathbb{R}^{p\times r}$. Output redundant systems, which is a subset of dynamical control systems given in (1), satisfy the following rank condition

$$\text{rank}(C) < p \tag{2}$$

and arise in sparse sensor applications[6], topologies used in Multi Agenr Systems(MAS)[4] and distributed networks [9]. These systems are named Strictly Output Redundant(SOR) system according to [8].

2.2 The Luenberger Observer

The Luenberger Observer is defined as,

$$\dot{\hat{x}} = A\hat{x} + L(y - \hat{y}),\ \hat{y} = C\hat{x} \tag{3}$$

where $L \in \mathbb{R}^{n\times p}$ is the observer gain. Assuming (A, C)-pair observable, the error dynamics are obtained for the error $e \triangleq x - \hat{x}$ as,

$$\dot{e} = (A - LC)e + (B_w - LD_w)w \tag{4}$$

The $\mathcal{H}_\infty$ optimal observer for objective function $z = C_z e$ is designed using the following LMI problem[2],

$$\min_{Y,P}(\gamma) \quad \text{s.t.}$$
$$\begin{bmatrix} (PA - YC) + \star & PB_w - YD_w & C_z^T \\ \star & -\gamma I & 0 \\ \star & 0 & -\gamma I \end{bmatrix} \prec 0 \tag{5}$$
$$P \succ 0$$

where the observer gain is recovered with $L = P^{-1}Y$.

2.3 Miscellaneous

The null space of a matrix X is denoted by $\mathcal{N}(X)$ where the columns space of X is denoted by $\mathcal{R}(X)$. The Moore-Penrose pseudo-inverse is $(.)^\dagger$ and defined as,

$$X^\dagger = X^T(XX^T)^{-1} \tag{6}$$

3 Projection Filtering

In any linear dynamical system with linear output mapping where output redundancy is present, the structure can be exploited to separate the output into its components. By using Gauge-Invariance principle a projection operator can be chosen to truncate some of the subcomponents of the output mapping to isolate the component under interest.

3.1 The Projection

Let X be a matrix and $X^\perp = \mathcal{N}(X^T)$ be its orthogonal representation, to separate the linear mapping

$$X + X^\perp \tag{7}$$

into its components, the following projection is proposed,

$$\Pi^X \triangleq I - X(X^TX)^{-1}X^T \tag{8}$$

which satisfies,

$$\Pi^X X = 0 \tag{9}$$

and

$$\Pi^X X^\perp = X^\perp \tag{10}$$

where $\Pi^X \in \mathbb{R}^{s\times s}$. In order to result in a meaningful projection, the output redundancy dependency defined in (2) for this projection is defined as,

$$\text{rank}(X) < s \tag{11}$$

The projection operator can be applied to the measured output defined in (1) to eliminate x or w. To eliminate x the projection Π^C is used as follows,

$$\Pi^C y = \Pi^C D_w w \tag{12}$$

and similarly, w is eliminated as follows,

$$\Pi^{D_w} y = \Pi^{D_w} Cx \tag{13}$$

In (12) the projection eliminates the state dependent component of the output, isolating the exogenous component establishing state invariance. The exogenous input can be recovered under the following condition,

$$\mathcal{R}(D_w) \not\subseteq \mathcal{R}(C) \tag{14}$$

so that

$$\Pi^C D_w \neq 0 \tag{15}$$

with the least square estimator,

$$\hat{\omega} = (\Pi^C D_w)^\dagger (\Pi^C y) \tag{16}$$

In a similar manner the state can be recovered using,

$$\hat{x} = (\Pi^{D_w})^\dagger (\Pi^{D_w} y) \tag{17}$$

Thus, both the state and disturbance signals are reconstructed rather than being estimated, exploiting the structural advantage of the output redundancy of systems defined with (1). As it is illustrated in the next section, using the projection filtering, the measured output is decomposed into its state and disturbance components algebraically, which shows that the projection filtering eliminates the requirement of individual observers to estimate the states and disturbances of such systems.

4 Numerical Examples

A comparison between the dynamic state estimation of a Luenberger observer and the algebraic recovery of both state and disturbance of the projection filtering method is given in Example 1. Noise is added in Example 2, in order to showcase that the projection filtering method is able to handle noise in measurements, by the cost of an additional sensor.

4.1 Example 1

The electrical system given in[8] with parameters $C_t = 2.2\text{mF}$, $L_t = 1.8\text{mH}$ and $R_t = 0.2\Omega$ is defined as the following state space model,

$$A = \begin{bmatrix} 0 & \frac{1}{C_t} \\ -\frac{1}{L_t} & -\frac{-R_t}{L_t} \end{bmatrix}, \; B_w = \begin{bmatrix} -\frac{1}{C_t} \\ 0 \end{bmatrix}, \; C = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{18}$$

where output redundancy is created by replicating the sensor measuring first state. The exogenous input matrix is constructed as follows,

$$D_w = C^\perp + C \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.7071 \\ 0.2929 \\ 1.0000 \end{bmatrix} \tag{19}$$

A Luenberger observer is designed solving the LMI given in (5) and is obtained as,

$$L = \begin{bmatrix} -326.1332 & 366.9908 & -5.2903 \\ -33.4833 & 28.1920 & 48.9023 \end{bmatrix} \tag{20}$$

noting that $B_w - LD_w = 0$ is satisfied, hence the state estimation is not affected by the exogenous input. The estimated state can be used to obtain the exogenous input by utilizing,

$$\hat{w} = D_w^\dagger (y - Cx) \tag{21}$$

On the other hand, using the projection filtering the projection matrix given by (8) is calculated for C as,

$$\Pi^C = \begin{bmatrix} 0.5 & -0.5 & 0 \\ -0.5 & 0.5 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{22}$$

and for D_w as,

$$\Pi^{D_w} = \begin{bmatrix} 0.2714 & -0.1250 & -0.4268 \\ -0.1250 & 0.9786 & -0.0732 \\ -0.4268 & -0.0732 & 0.7500 \end{bmatrix} \tag{23}$$

The disturbance and the states are filtered using (16) and (17), respectively. The state estimation and the projection state filtering is compared in Figure 1.

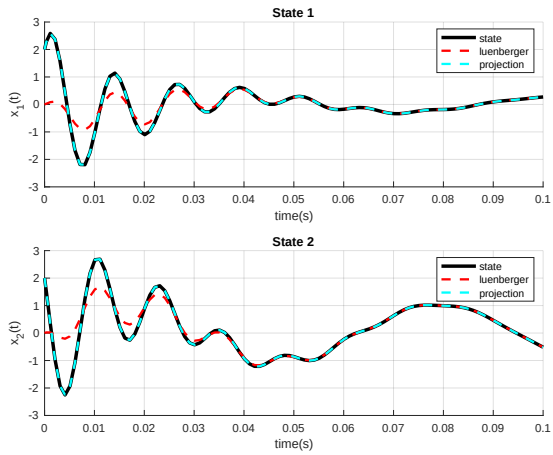

Fig. 1. Comparison of the Luenberger observer state estimation versus projection filtering state reconstruction.

The estimation process of the Luenberger observer takes 0.04 seconds due to the fact that the observer is a dynamic observer, whereas the projection filtering directly matches the original system states at $t = 0$ seconds as

illustrated in Figure 1. It is possible to decrease the estimation duration of the observer by speeding up the observer eigenvalues, but in return the observer gain and therefore noise amplification would increase.

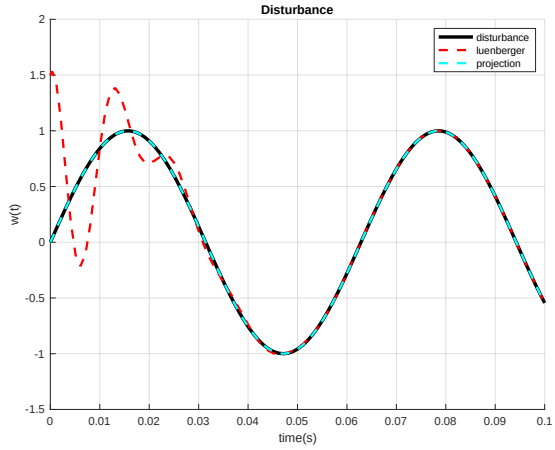

Fig. 2. Estimated disturbance using Luenberger Observer versus reconstructed disturbance of the projection filtering

The Luenberger observer is a state estimator, but using the linear mapping of the output, it is possible to reconstruct the disturbance assuming the lack of noise in the measured output. It also possible to employ a Disturbance Observer, but then it is necessary to also assume a certain model or add an assumption about the disturbance. This extra information about the disturbance is not necessary for the projection filtering method. As illustrated in Figure 2, the reconstruction of the disturbance using the Luenberger observer state matches the disturbance only in the steady state, since the state estimation gets corrected during the transients. But, the projection filtering method results in an exact match, due to its static approach.

4.2 Example 2

In most practical applications noise is present in measured outputs as given in,

$$y = Cx + D_w w + D_n n \tag{24}$$

For illustrative purposes the following output equation is defined

$$y = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} x(t) + \begin{bmatrix} 2.2060 \\ 2.3970 \\ 1.3015 \\ 1.3015 \end{bmatrix} w(t) + \begin{bmatrix} 0.1960 \\ 0.7382 \\ 0.6697 \\ 0.5265 \end{bmatrix} n(t) \tag{25}$$

where $w(t)$ is disturbance, $n(t)$ is noise and $x(t)$, which represent some system states and are arbitrarily chosen as,

$$x(t) = \begin{bmatrix} \sin(t) \\ \cos(t) \end{bmatrix} \tag{26}$$

In order to be able to recover both the states and the disturbance in such a system, the number of sensors for measurement needs to increase due to the condition given in (11). The disturbance matrix is defined using

$$D_w = C \begin{bmatrix} 1 \\ 1 \end{bmatrix} + C^{\perp} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \tag{27}$$

and the noise matrix D_n is chosen as a random matrix.

The output mapping in (24) is rewritten as,

$$y = Cx + \begin{bmatrix} D_w & D_n \end{bmatrix} \begin{bmatrix} w \\ n \end{bmatrix} \tag{28}$$

and the projection defined in (8) to recover the state information $x(t)$ can be calculated using,

$$\Pi^{wn} = I - \begin{bmatrix} D_w & D_n \end{bmatrix} \left(\begin{bmatrix} D_w \\ D_n \end{bmatrix} \begin{bmatrix} D_w & D_n \end{bmatrix} \right)^{-1} \begin{bmatrix} D_w \\ D_n \end{bmatrix} \tag{29}$$

and is determined as

$$\Pi^{wn} = \begin{bmatrix} 0.6224 & -0.2581 & -0.3034 & -0.2763 \\ -0.2581 & 0.1118 & 0.1698 & 0.0618 \\ -0.3034 & 0.1698 & 0.5564 & -0.3549 \\ -0.2763 & 0.0618 & -0.3549 & 0.7094 \end{bmatrix} \tag{30}$$

Then the state $x(t)$ is recovered using,

$$\hat{x} = (\Pi^{wn} C)^{\dagger} (\Pi^{wn} y) \tag{31}$$

The measured output for each sensor with noise is depicted in Figure 3. The output redundancy can be observed by comparing third output and fourth output in Figure 3. It can also be seen that the projection filtering method is able to reconstruct both states under noisy measurements.

The disturbance is recovered using the projection on $\begin{bmatrix} C & D_n \end{bmatrix}$ as,

$$\Pi^{xn} = \begin{bmatrix} 0.7254 & -0.3627 & 0.1507 & 0.2120 \\ -0.3627 & 0.1813 & -0.0754 & -0.1060 \\ 0.1507 & -0.0754 & 0.0313 & 0.0440 \\ 0.2120 & -0.1060 & 0.0440 & 0.0619 \end{bmatrix} \tag{32}$$

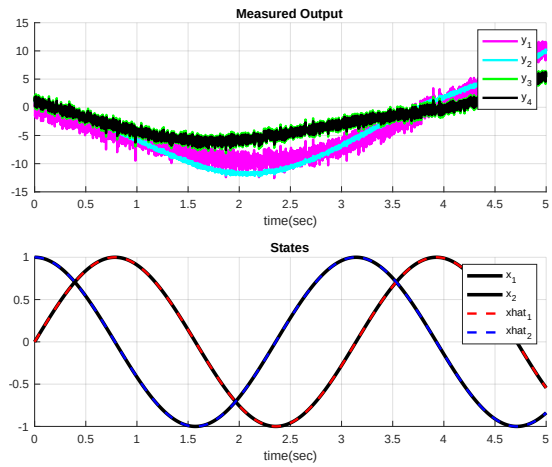

Fig. 3. The measured outputs(y_1, y_2, y_3, y_4) and the states(x_1, x_2) versus the projection filtering method recovered states($\hat{x}_1$, $\hat{x}_2$)

and is plotted in Figure 4. It is observed that disturbance recovery works in a similar fashion compared to state recovery in Figure 3.

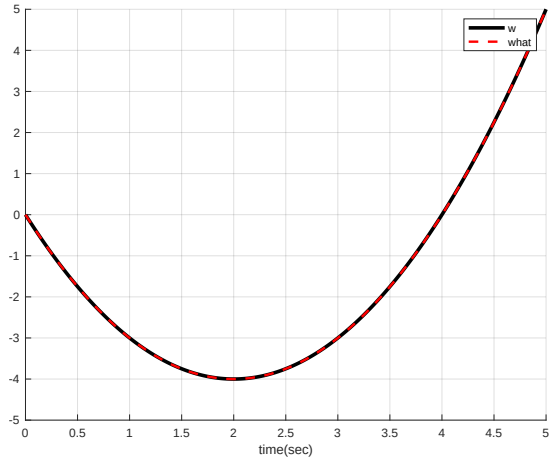

Fig. 4. Disturbance recovered by projection filtering versus actual disturbance

Throughout the examples given in this section, it has been established that, the projection filtering method exploits the output mapping using a projection matrix and recovers the states and the disturbance of a system directly. Whereas, the Luenberger observer, however fast dynamics it possesses, estimates the states of a system, due to its dynamic structure. Assuming no information about disturbance, one can also back-calculate the disturbance in a system with the Luenberger state estimates. Due to the dynamics, the disturbance and state estimations are valid for steady-state. But projection filtering gives valid results for both transient and steady-state regime. Additionally, projection filtering can handle noise, if necessary amount of sensors are present.

5 Conclusion and Future Work

For estimation of states of output redundant systems, the Luenberger observer is a classical tool. This extends to disturbance estimation, if a model or additional assumptions are made for the disturbance.

However, it has been shown that, for output redundant systems, a projection filtering method can be used to both reconstruct the states and disturbances. The PFM is based on the gauge-invariance principle, where the invariance can be chosen to isolate components for recovery under appropriate transformations, which then in turn is used to separate the linear mapping in the output. It has been noted that the separation is a static process, rather than an estimation process. It has been observed that, the static nature proves to be an important advantage, since this can be achieved with high gain Luenberger observers, but with a cost of higher noise amplification.

It is also shown that, the PFM can be applied for outputs where noise is present. It is observed that the PFM is an alternative to classical observers for output redundant systems.

A natural extension of this work is to apply PFM to systems with nonlinear dynamics and output mappings, etc.

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